Cambridge Assessment International Education

Cambridge International A Level

MATHEMATICS **9709/32**

Paper 3 Pure Mathematics 3 **March 2021**

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the March 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

© UCLES 2021

[Turn over

PUBLISHED

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:
Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:
Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:
Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:
Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:
Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

© UCLES 2021

Mathematics Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

© UCLES 2021

PUBLISHED

Mark Scheme Notes

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

DM or **DB** When a part of a question has two or more 'method' steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

FT Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.

- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
- For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
- The total number of marks available for each question is shown at the bottom of the Marks column.
- Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
- Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

© UCLES 2021

Abbreviations

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no 'follow through' from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

© UCLES 2021

Question	Answer	Marks	Guidance
1	Use law of the logarithm of a product or power	**M1**	
	Obtain a correct equation free of logarithms, e.g. $3\left(x^3-3\right)=x^3$	**A1**	
	Obtain $x = 1.65$	**A1**	
		3	

Question	Answer	Marks	Guidance
2	Substitute $x = -2$, equate result to zero and obtain a correct equation, e.g. $-8a + 20 + 8 + b = 0$	**B1**	
	Substitute $x = -1$ and equate result to 2	**M1**	
	Obtain a correct equation, e.g. $-a + 5 + 4 + b = 2$	**A1**	
	Solve for a or for b	**M1**	
	Obtain $a = 3$ and $b = -4$	**A1**	
		5	

© UCLES 2021

Question	Answer	Marks	Guidance
3	Use correct trig formulae to obtain an equation in $\tan x$	***M1**	
	Using $\tan 45° = 1$, obtain a horizontal equation in $\tan x$ in any form	**DM1**	
	Reduce the equation to $\tan^2 x + \tan x - 1 = 0$, or 3-term equivalent	**A1**	
	Solve a 3-term quadratic in tan x, for x	**M1**	
	Obtain answer, e.g. $x = 31.7°$	**A1**	
	Obtain second answer, e.g. $x = 121.7°$, and no other in the interval	**A1**	Ignore answers outside the given interval.
		6	

Question	Answer	Marks	Guidance
4(a)	Separate variables correctly and attempt integration of at least one side	**M1**	
	Obtain term $\ln y$	**A1**	
	Obtain term of the form $\pm\ln(1-\cos x)$	**M1**	
	Obtain term $\ln(1-\cos x)$	**A1**	
	Use $x = \pi$, $y = 4$ to evaluate a constant, or as limits, in a solution containing terms of the form $a\ln y$ and $b\ln(1-\cos x)$	**M1**	
	Obtain final answer $y = 2(1-\cos x)$	**A1**	OE
		6	

Question	Answer	Marks	Guidance
4(b)	Show a correct graph for $0 < x < 2\pi$ with the maximum at $x = \pi$	**B1 FT**	The FT is for graphs of the form $y = a(1 - \cos x)$, where a is positive.
		1	

Question	Answer	Marks	Guidance
5(a)	State $R = \sqrt{11}$	**B1**	
	Use trig formulae to find α	**M1**	
	Obtain $\alpha = 37.09°$	**A1**	
		3	
5(b)	Evaluate $\sin^{-1}\left(\frac{1}{\sqrt{11}}\right)$ to at least 2 dp (17.5484°)	**B1 FT**	The FT is on R.
	Use correct method to find a value of θ in the interval	**M1**	
	Obtain answer, e.g. 62.7°	**A1**	
	Use a correct method to obtain a second answer	**M1**	
	Obtain second answer, e.g. 170.2° , and no other in the interval	**A1**	Ignore answers outside the given interval.
		5	

© UCLES 2021

Question	Answer	Marks	Guidance
6(a)	Carry out a relevant method to determine constants A and B such that $\frac{5a}{(2x-a)(3a-x)} = \frac{A}{2x-a} + \frac{B}{3a-x}$	**M1**	
	Obtain $A = 2$	**A1**	
	Obtain $B = 1$	**A1**	
		3	
6(b)	Integrate and obtain terms $\ln(2x-a) - \ln(3a-x)$	**B1 FT** **B1 FT**	The FT is on the values of A and B.
	Substitute limits correctly in a solution containing terms of the form $b\ln(2x-a)$ and $c\ln(3a-x)$, where $bc \neq 0$	**M1**	
	Obtain the given answer showing full and correct working	**A1**	
		4	

© UCLES 2021

Question	Answer	Marks	Guidance
7(a)	Express general point of a line in component form, e.g. $(1 + 2s, 3 – s, 2 + 3s)$ or $(2 + t, 1 – t, 4 + 4t)$	**B1**	
	Equate at least two pairs of components and solve for s or for t	**M1**	
	Obtain correct answer for s or for t (possible answers are $–1$, 6, $\frac{2}{5}$ for s and $–3$, 4, $-\frac{1}{5}$ for t)	**A1**	
	Verify that all three component equations are not satisfied	**A1**	
	Show that the lines are not parallel and are thus skew	**A1**	
		5	
7(b)	Carry out correct process for evaluating the scalar product of the direction vectors	**M1**	
	Using the correct process for the moduli, divide the scalar product by the product of the moduli and evaluate the inverse cosine of the result	**M1**	
	Obtain answer $19.1°$ or 0.333 radians	**A1**	
		3	

Question	Answer	Marks	Guidance
8(a)	Multiply numerator and denominator by 3 – i	**M1**	OE
	Obtain numerator – 10 + 10i or denominator 10	**A1**	
	Obtain final answer – 1 + i	**A1**	
		3	
8(b)	State or imply $r = \sqrt{2}$	**B1 FT**	
	State or imply that $\theta = \frac{3}{4}\pi$	**B1 FT**	
		2	
8(c)	State that *OA* and *BC* are parallel	**B1**	
	State that $BC = 2OA$	**B1**	
		2	

© UCLES 2021

Question	Answer	Marks	Guidance
8(d)	Use angle $AOB = \arg u - \arg v = \arg\dfrac{u}{v}$	**M1**	
	Obtain the given answer	**A1**	
	Alternative method for question 8(d)		
	Obtain tan AOB from gradients of OA and OB and the $\tan(A \pm B)$ formula	**M1**	
	Obtain the given answer	**A1**	
	Alternative method for question 8(d)		
	Obtain $\cos AOB$ by using the cosine rule or a scalar product	**M1**	
	Obtain the given answer	**A1**	
		2	

Question	Answer	Marks	Guidance
9(a)	Calculate the values of a relevant expression or pair of expressions at $x = 1$ and $x = 1.5$	**M1**	
	Complete the argument correctly with correct calculated values	**A1**	
		2	
9(b)	Use the iterative formula $x_{n+1} = \dfrac{e^{2x_n}+1}{e^{2x_n}-1}$, or equivalent, correctly at least once	**M1**	
	Obtain final answer 1.20	**A1**	
	Show sufficient iterations to 4 dp to justify 1.20 to 2 dp, or show there is a sign change in the interval (1.195,1.205)	**A1**	
		3	

Question	Answer	Marks	Guidance
9(c)	Use quotient rule	**M1**	
	Obtain correct derivative in any form	**A1**	
	Equate derivative to – 8 and obtain a quadratic in e^{2x}	**M1**	
	Obtain $2\left(e^{2x}\right)^2 - 5e^{2x} + 2 = 0$	**A1**	OE
	Solve a 3-term quadratic in e^{2x} for x	**M1**	
	Obtain answer $x = \frac{1}{2}\ln 2$, or exact equivalent, only	**A1**	
	Alternative method for question 9(c)		
	Use quotient rule	**M1**	
	Obtain correct derivative in any form	**A1**	
	Equate derivative to – 8, take square roots and obtain a quadratic in e^{x}	**M1**	
	Obtain $\sqrt{2}e^{2x} - e^{x} - \sqrt{2} = 0$	**A1**	OE
	Solve a 3-term quadratic in e^{x} for x	**M1**	
	Obtain answer $x = \frac{1}{2}\ln 2$, or exact equivalent, only	**A1**	
		6	

© UCLES 2021

Question	Answer	Marks	Guidance
10(a)	State or imply du = cos x dx	**B1**	
	Using double angle formula for sin2x and Pythagoras, express integral in terms of u and du.	**M1**	
	Obtain integral $\int 2\left(u-u^3\right)\mathrm{d}u$	**A1**	OE
	Use limits $u = 0$ and $u = 1$ in an integral of the form au^2+bu^4, where $ab \neq 0$	**M1**	$a + b$ or $a + b - 0$ $\left(a=1 \text{ and } b=-\frac{1}{2}\right)$
	Obtain answer $\frac{1}{2}$	**A1**	
		5	
10(b)	Use product rule	**M1**	
	Obtain correct derivative in any form	**A1**	
	Equate derivative to zero and use a double angle formula	***M1**	
	Obtain an equation in one trig variable	**DM1**	
	Obtain $4\sin^2 x=1$, $4\cos^2 x=3$ or $3\tan^2 x=1$	**A1**	
	Obtain answer $x=\frac{1}{6}\pi$	**A1**	
		6	

© UCLES 2021